

Optimization strategy based on robust model predictive control for RES-CCHP system under multiple uncertainties

Xing Dong, Chenghui Zhang^{*}, Bo Sun

School of Control Science and Engineering, Shandong University, Jingshi Road 17923, Jinan 250061, China

HIGHLIGHTS

- An integrated framework is constructed to coordinate prediction and optimization.
- An optimization strategy based on robust model predictive control is proposed.
- A min-max rolling optimization model of the RES-CCHP system is presented.
- Source-load probability distribution is modeled by the Gaussian process regression.

ARTICLE INFO

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Multiple uncertainties
Uncertain rolling optimization framework
Robust model predictive control

ABSTRACT

The economic operation of renewable energy sources (RES) integrated with a combined cooling, heating, and power system significantly improves energy utilization and reduces environmental crises. However, multiple uncertainties in RES generation and load consumption predictions cause an imbalance between supply and demand, negatively impacting system efficiency and economics. To address this limitation, an integrated uncertainty rolling optimization framework for combining the probability distribution of prediction and uncertainty optimization is established. Under this framework, an optimization strategy based on a robust model predictive control is proposed for handling the multiple uncertainties of source-load. Here, the probability distributions of RES generation and load consumption predictions are determined using the Gaussian process regression method, and a minimum-maximum rolling optimization model is developed. Under the uncertain scenarios of RES generation and load consumption, the optimization model is converted into a tractable form to obtain a robust schedule that minimizes operating costs. The over-conservatism of robust optimization can be mitigated by adjusting the uncertainty budget. Many case studies are further conducted to confirm the effectiveness of the proposed method. Results show that the operating costs decreased by 11.5% compared with the traditional model predictive control strategies in an uncertain scenario.

1. Introduction

The supply and consumption of energy are the basis of rapid economic development [1]. However, the depletion of energy sources and climate issues have attracted considerable attention globally. For sustainable economic development, it is essential to increase the penetration of renewable energy sources (RES) [2]. Incorporating RES into combined cooling, heating, and power (RES-CCHP) systems can improve primary energy utilization by combining many energy sources [3]. As the RES-CCHP system has a high energy efficiency, it has been extensively used to provide electricity, heating, and cooling for campuses [4] and commercial buildings [5]. However, RES generation and load

consumption fluctuate considerably (source-load uncertainties), seriously affecting the operation performance of RES-CCHP systems [6]. The problem usually needs to be solved using advanced operation optimization methods [7]. The implementation of operation optimization primarily includes two parts: (1) prediction of RES generation and loads consumption (source-load prediction) and (2) modeling of the optimization strategy (OS). The literature review is discussed separately for these two parts in the subsequent paragraphs.

First, source-load prediction is the input to the OS, and accurate prediction is conducive to improving the optimization effect. Currently, prediction methods (PMs) have been increasingly studied. In Ref. [8], the general trend of wind energy is predicted using artificial intelligence technology and data fusion algorithm through various neural networks.

^{*} Corresponding author.

E-mail address: zchsdu@163.com (C. Zhang).

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Nomenclature		Q	length of prediction horizon
<i>Abbreviations</i>		U	optimization variables
RES	renewable energy sources	SV	state variables
CCHP	combined cooling, heating, and power	W	error set
ROF	rolling optimization framework	τ	self-discharge coefficient
PM	prediction method	f	underlying function
OS	optimization strategy	gp	Gaussian process
GPR	Gaussian process regression	m	mean function
RMPC	robust model predictive control	k	covariance function
MILP	mixed-integer linear programming	I	identity matrix
CNY	Chinese Yuan	λ	hyperparameters
MG	main grid	ε	noise
PGU	power generation units	σ	standard deviation
WT	wind turbines	UB	upper bound
GB	gas boiler	η	efficiency
AC	absorption chiller	ξ	state of ESU and HSU
EC	electric chiller	δ	binary sign of prediction
ESU	electricity storage units	<i>Subscripts</i>	
HSU	heat storage units	pgu	power generation units
HRD	heat recovery devices	grid	main grid
HE	heat exchanger	es	electricity storage
<i>Symbols</i>		hs	heat storage
E	electricity demand (kW)	ec	electric chiller
C	cooling demand (kW)	ac	absorption chiller
H	heat demand (kW)	gb	gas boiler
G	energy consumption (kW)	he	heat exchanger
Π	uncertainty budget	res	renewable energy sources
COP	coefficient of performance	chr/dis	energy charge/discharge
J	objective function	l	load
S	training set	hr	heat recovery
X	input data of the GPR model	err	error
Y	output data of the GPR model	sta	state of energy storage units
M	length of optimization horizon	min-max	minimum-maximum

In Ref. [9], a deep learning model with a dynamic scene cross-generation algorithm of photovoltaic power generation was established to improve prediction accuracy. Concerning the load forecast, to improve the stability of the forecasting model, a short-term power-load-forecasting method, including error correction based on dynamic mode decomposition, has been proposed [10]. In Ref. [11], a novel dense average connection structure was proposed to realize a short-term load-forecasting, and an integrated method was applied to reduce the forecast deviation. For predicting cooling, heating, and power load in CCHP systems, a forecasting model based on an autoregressive moving average with exogenous inputs has been proposed [12]. The abovementioned forecasting methods only determine the forecast value, whereas the forecasting model based on Gaussian process regression (GPR) can predict the source-load probability distribution. A multitask GPR model [13] and a hybrid model that combines long short-term memory with GPR [14] have been proposed to predict renewable energy generation. To solve the problem of unreasonable training set selection, a short-term load-forecasting model based on the combination of density value clustering and GPR has been established [15].

Second, OSs are important for improving system operations. Initially, scholars mainly studied the day-ahead OSs to develop efficient and economical source-load matching plans. In Ref. [16], an OS considering multiple objectives was established, and a new electrical load-tracking strategy was proposed to improve energy utilization as per the energy requirements of buildings. In Ref. [17], to promote the low-carbon operation of a CCHP system, a bilevel carbon pricing OS was developed, and a bisection method was proposed to obtain effective

incentives. In Ref. [18], a trilayer OS of multi-time-space scale was established to guarantee the energy supply and load demand balance. In Ref. [19], to improve the system's economy, a dynamic model of a CCHP system pipe network was proposed, and a day-ahead OS considering energy storage units was proposed. In Ref. [20], an OS considering multi-objective and nonlinear constraints was developed, and a gray wolf solution algorithm was proposed, significantly improving the energy efficiency. However, the intermittent nature of RES and the randomness of the customer loads negatively affect the operation of the RES-CCHP system [21]. To address this limitation, in Ref. [22], a day-ahead OS, including load and RES uncertainty, was established, and benefiting from the advantages of digital twins, the operating costs of the CCHP system were reduced. To guarantee the stable operation of a CCHP system with uncertainty in the RES power generation, an adaptive robust OS has been established [23]. In Ref. [24], a multiple thermal inertia model of a CCHP system was studied, and a robust OS containing the thermal inertia uncertainty was proposed, which increased the thermal network's heat storage capacity. In Ref. [25], a CCHP model, including an energy recovery system, was developed, and a two-stage robust OS considering the RES uncertainty and water inrush was adopted to fully improve the energy recovery, economic benefits, and operational robustness. In Ref. [26], robustness was introduced into a genetic algorithm, and a multi-objective robust OS of the CCHP system was established to evaluate the system's environmental impact. In Ref. [27], to ensure the stable operation of a CCHP microgrid system in case of a small number of source-load samples, an adaptive kernel density estimator was used to predict the source-load probability

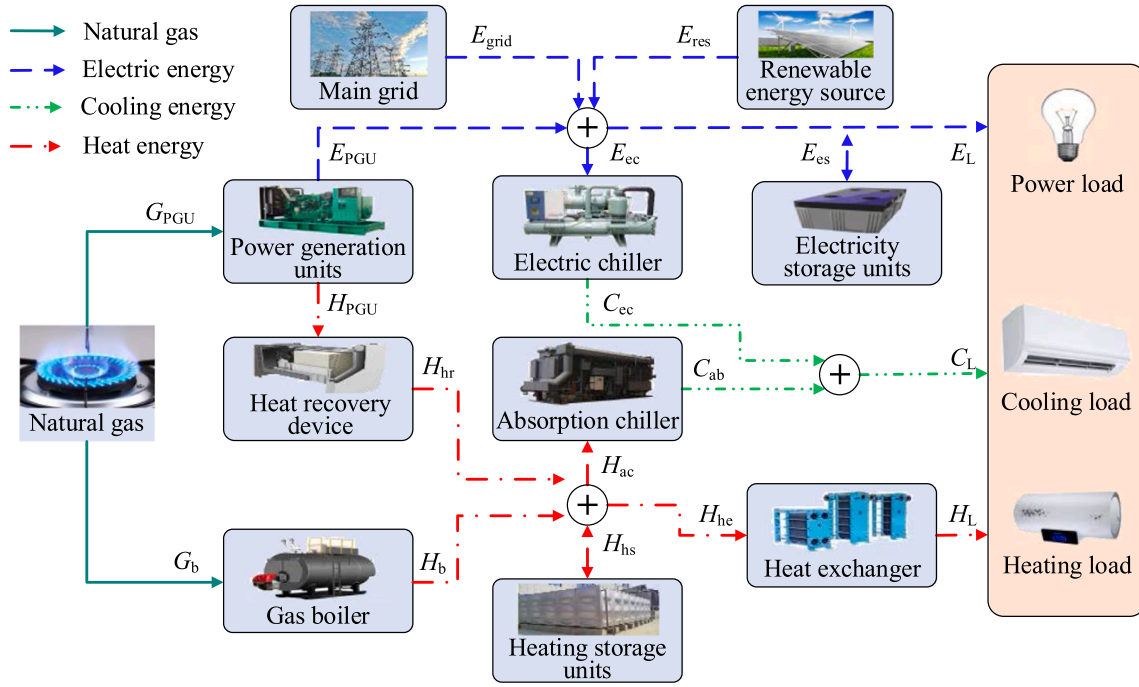


Fig. 1. Configuration and energy flows of RES-CCHP system.

distribution, and a day-ahead uncertainty OS based on probabilistic chance constraints was proposed. The foregoing day-ahead OSs have made significant contributions to the operation of CCHP systems. However, with the increasing uncertainty of source and load, these day-ahead OSs cannot adjust the schedules on an intraday basis in response to rapid changes in RES and customer loads, which affects the intraday's economical and efficient operation of the system.

To address the defects of the day-ahead OSs, the rolling optimization technology based on the model predictive control (MPC) method has been extensively studied [28]. In the MPC-based optimization, the system's future state can be obtained using a predictive model, and a rolling OS determines the output sequence in a limited time horizon. Moreover, feedback adjustments can be implemented as per the current status at each optimization interval. For the economic dispatch of a CCHP microgrids system, in Ref. [29], an OS combining dynamic time interval and MPC was proposed to optimize the system's operating costs. In Ref. [30], an MPC model based on affine technology was proposed, which provided a cost-effective solution. In Ref. [31], an OS combining a day-ahead distributionally robust chance-constrained optimization model with an MPC-based rolling optimization model was established to trade off economics and operational risks. To realize the real-time operation of a CCHP system, an OS combining random day-ahead optimization and real-time MPC was proposed in [32]. In Ref. [33], a two-stage coordinated control method was proposed, and in the economic-dispatching stage, MPC was used to schedule the system's operation. In Ref. [34], an MPC framework was applied in a scheduling method for a CCHP system, including rolling and feedback correction parts and it can better achieve source-load matching. The above MPC-based OSs can adapt to the rapid changes in the source-load and resolve the impact of the day-ahead prediction error on the system operation. However, the intraday prediction error caused by source-load uncertainty has not been considered in the above MPC-based OS modeling, which is un conducive to enhancing the abilities of the system's intraday operations in resisting multiple uncertain disturbances.

Due to the existence of intraday prediction errors, existing MPC-based OSs cannot be directly deployed. In addition, dealing with intraday prediction errors requires considering time scale differences between multiple energy forms, which also increases the complexity of

the OS. Therefore, to improve the robust and economic performance of the intraday operations of the RES-CCHP system under the multiple uncertainties of RES generation and load consumption, this study proposes an OS based on robust MPC (RMPC) and considers the uncertainty characteristics of source-load in the optimization modeling. The primary contributions of this study are summarized as follows:

- (1) An integrated uncertainty rolling optimization framework (ROF) that includes the probability distribution of prediction and uncertainty rolling optimization is established. Specifically, under the framework, forecasting and optimization are implemented on a rolling time horizon, and the probability distribution of prediction uses different time scales and can be corrected as per the system's current state.
- (2) Based on the uncertainty ROF, an RMPC-based OS involving GPR-based source-load prediction and minimum-maximum (min-max) rolling optimization is proposed. The proposed strategy applies the RMPC theory to the optimal operation of the RES-CCHP system, and the uncertainty budget parameters are introduced to adjust the robust optimization conservativeness.
- (3) In a source-load uncertainty scenario, the min-max optimization model is converted into a tractable form for solving. The case study confirms the advantages of the proposed OS by comparing it with traditional MPC-based and day-ahead robust OSs.

The rest of this study is organized as follows: Section 2 describes the RES-CCHP system structure and the energy flows. Section 3 introduces the integrated uncertainty ROF and the RMPC-based OS. Section 4 introduces the detailed solution process of the min-max rolling optimization model. Section 5 presents the case study results, and the last section concludes the study.

2. System structure and energy flow analysis

2.1. RES-CCHP system structure

Compared with the traditional single energy flow system, the RES-CCHP system primarily comprises three energy flows: electrical,

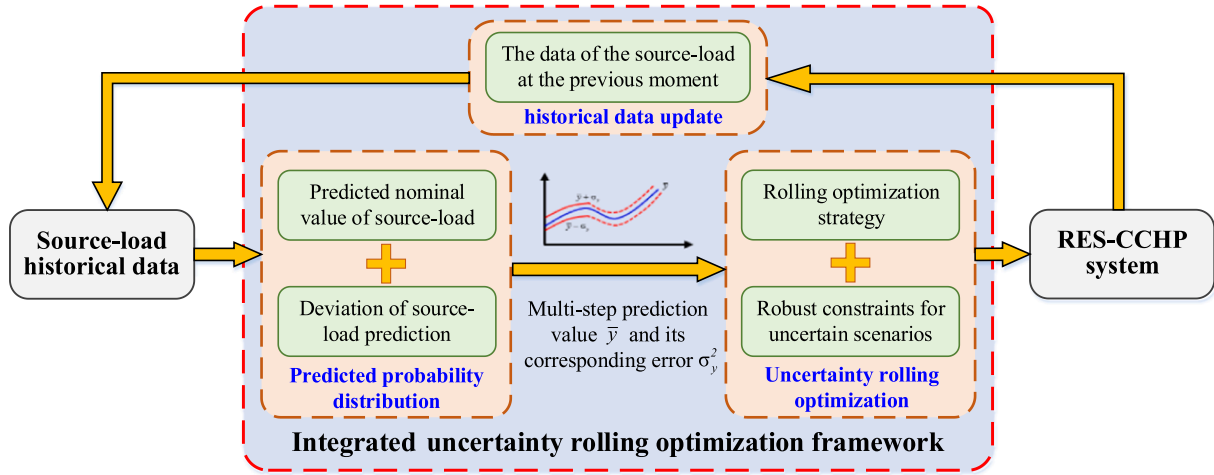


Fig. 2. Organizational structure of integrated uncertainty ROF.

cooling, and heat energy flow. The cascade utilization and complementary coupling of energy can be realized by coordinating the operation of the three energy flows, and the efficient utilization of energy can be guaranteed. Moreover, the system is attached to the main grid (MG), which can improve the economy and flexibility of system operation through interaction with the MG. Fig. 1 shows the configuration and the energy flows of the RES-CCHP system.

Particularly, the electrical energy flow comprises four components: MG, power generation units (PGU), wind turbines (WT), and electricity storage units (ESU). The cooling energy flow comprises an electric chiller (EC) and an absorption chiller (AC). The heat energy flow primarily comprises the gas boiler (GB), heat recovery devices (HRD), heat exchanger (HE), and heat storage units (HSU). The primary functions of these components are listed as follows: (1) PGU and GB consume natural gas to generate electricity and heat, respectively. (2) EC and AC convert electricity and heat energy into cooling energy, respectively. (3) The waste heat of the PGU, including flue gas waste heat and jacket water heat, can be recovered through HR. (4) The heat energy is converted by HE to meet the heating demand. (5) The energy storage units (ESU and HSU) can provide more operational flexibility for system operation.

2.2. Energy flow analysis

The supply and demand balance in the RES-CCHP system should be guaranteed during the operational scheduling of electricity, heating, and cooling. To avoid energy flow imbalance caused by the uncertainty of renewable energy and load, the error factor of predictions (including RES generation and load consumption predictions) was introduced in the energy flow analysis. The electrical, cooling, and heat energy flow balances at time (t) are described by Eqs. (1)–(3), respectively.

$$\begin{aligned} E_{\text{pgu}}(t) + E_{\text{grid}}(t) + E_{\text{es,dis}}(t) + E_{\text{res}}(t) + E_{\text{res}}^{\text{err}}(t) \\ = E_{\text{ec}}(t) + E_{\text{es,chr}}(t) + E_{\text{h}}(t) + E_{\text{h}}^{\text{err}}(t) \end{aligned} \quad (1)$$

$$C_{\text{ec}}(t) + C_{\text{ac}}(t) = C_{\text{l}}(t) + C_{\text{err}}(t) \quad (2)$$

$$H_{\text{hr}}(t) + H_{\text{gb}}(t) + H_{\text{hs,dis}}(t) = H_{\text{h}}(t)/\eta_{\text{he}} + H_{\text{ac}}(t) + H_{\text{hs,chr}}(t) + H_{\text{err}}(t) \quad (3)$$

In Eq. (1), E_{pgu} represents the PGU power output, E_{grid} represents the power purchased from the MG, $E_{\text{es,dis}}$ and $E_{\text{es,chr}}$ are the discharge and charge power of the ESU, E_{res} and $E_{\text{res}}^{\text{err}}$ are the power generation and prediction error of the RES, E_{ec} represents the electric power consumed by the EC, and E_{h} and $E_{\text{h}}^{\text{err}}$ represent the electrical load and its prediction error, respectively. In Eq. (2), C_{ec} and C_{ac} represent the cooling energies provided by the EC and AC, respectively. Furthermore, C_{l} and C_{err} represent the cooling load and the prediction error of the cooling load,

respectively. In Eq. (3), H_{hr} represents the total waste heat of the PGU, H_{gb} is the heat output of the GB, $H_{\text{hs,dis}}$ and $H_{\text{hs,chr}}$ represent the heat output and input of the HSU, and η_{he} represents the heat conversion efficiency of the HE. H_{l} and H_{err} denote the heating load and its prediction error, respectively.

The PGU energy conversion in the electrical energy flow can be expressed as follows:

$$E_{\text{pgu}}(t) = G_{\text{pgu}}(t)\eta_{\text{pgu}} \quad (4)$$

$$H_{\text{hr}}(t) = \frac{E_{\text{pgu}}(t)}{\eta_{\text{pgu}}} (1 - \eta_{\text{pgu}})\eta_{\text{hr}} \quad (5)$$

where G_{pgu} represents the natural gas consumption of the PGU, and η_{pgu} and η_{hr} represent the PGU and HRU electrical and recovery efficiencies, respectively.

In the cooling energy flow, the AC and EC guarantee the cooling load demand of the users. Their power output can be described in Eqs. (6) and (7).

$$C_{\text{ec}}(t) = E_{\text{ec}}(t)COP_{\text{ec}} \quad (6)$$

$$C_{\text{ac}}(t) = H_{\text{ac}}(t)COP_{\text{ac}} \quad (7)$$

Here, H_{ac} is the heat consumed by the AC, and COP_{ec} and COP_{ac} denote the coefficient of performance (COP) values of EC and AC, respectively.

In practice, GB is often used in RES-CCHP systems to add flexibility. The heat energy conversion of the GB is expressed as:

$$H_{\text{gb}}(t) = G_{\text{gb}}(t)\eta_{\text{gb}} \quad (8)$$

where G_{gb} and η_{gb} are the natural gas consumption and the heat efficiency of the GB.

Energy storage units are critical components of RES-CCHP systems because they balance the supply and demand by regulating the intermittent RES and peak loads. The dynamic energy balance of the ESU is expressed as follows:

$$E_{\text{sta}}(t+1) = (1 - \tau_{\text{es}})E_{\text{sta}}(t) + E_{\text{es,chr}}(t)\eta_{\text{es,chr}} - E_{\text{es,dis}}(t)/\eta_{\text{es,dis}} \quad (9)$$

where E_{sta} represents the state of the ESU, τ_{es} represents the self-discharge coefficient of the ESU, and $\eta_{\text{es,chr}}$ and $\eta_{\text{es,dis}}$ are the energy charge and discharge efficiency of the ESU, respectively. The ESU constraints are presented as follows:

$$0 \leq E_{\text{sta}}(t) \leq E_{\text{sta,max}} \quad (10)$$

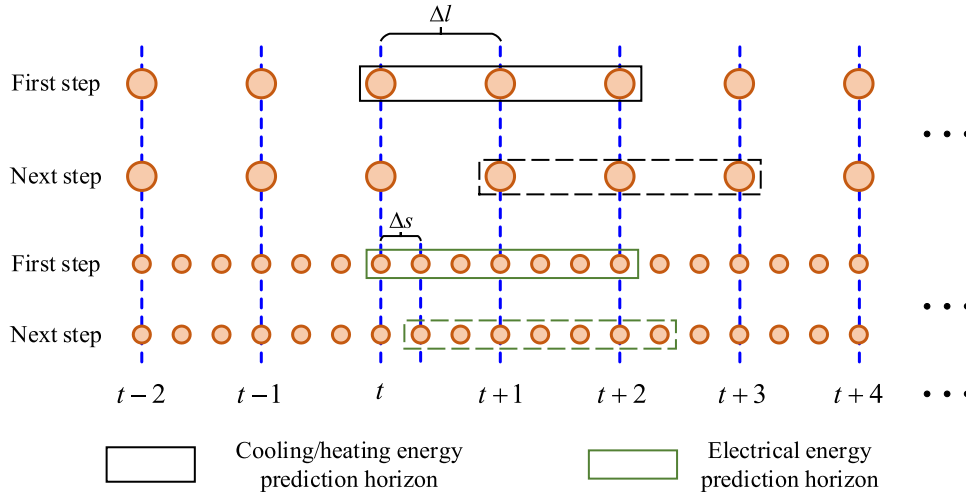


Fig. 3. Prediction of electrical, cooling, and heating in different time scales.

$$0 \leq E_{es,chr}(t) \leq \xi_{es,chr}(t) E_{es,chr,max} \quad (11)$$

$$0 \leq E_{es,dis}(t) \leq \xi_{es,dis}(t) E_{es,dis,max} \quad (12)$$

$$\xi_{es,chr}(t) + \xi_{es,dis}(t) \leq 1; \quad \xi_{es,chr}(t), \xi_{es,dis}(t) \in \{0, 1\} \quad (13)$$

where $E_{sta,max}$ denotes the maximum capacity of the ESU, and $E_{es,chr,max}$ and $E_{es,dis,max}$ represent the upper limits of the charge and discharge power, respectively. $\xi_{es,chr}$ and $\xi_{es,dis}$ represent the charge and discharge states.

Similar to the ESU, the corresponding dynamic energy balance and constraints of the HSU are formulated as follows:

$$H_{sta}(t+1) = (1 - \tau_{hs})H_{sta}(t) + H_{hs,chr}(t)\eta_{hs,chr} - H_{hs,dis}(t)/\eta_{hs,dis} \quad (14)$$

$$0 \leq H_{sta}(t) \leq H_{sta,max} \quad (15)$$

$$0 \leq H_{hs,chr}(t) \leq \xi_{hs,chr}(t) H_{hs,chr,max} \quad (16)$$

$$0 \leq H_{hs,dis}(t) \leq \xi_{hs,dis}(t) H_{hs,dis,max} \quad (17)$$

$$\xi_{hs,chr}(t) + \xi_{hs,dis}(t) \leq 1; \quad \xi_{hs,chr}(t), \xi_{hs,dis}(t) \in \{0, 1\} \quad (18)$$

where the definition of symbols and variables are similar to those in Eqs. (9)–(13), which are omitted here.

3. Uncertainty ROF and RMPC-based OS

3.1. Integrated uncertainty ROF

Herein, an integrated uncertainty ROF was established to explore the collaboration between PMs and uncertain OSs in the optimal scheduling of RES-CCHP systems. The integrated optimization framework comprises three parts (Fig. 2): (1) predicted probability distribution, (2) uncertainty rolling optimization, and (3) historical data update.

Under this framework, first, a prediction model of the probability distribution was constructed to predict the source-load and its corresponding error interval. Because the electrical load changes quickly and the cooling and heating loads change slowly, different time scales were used for the part of the predicted probability distribution. Then, the obtained forecast data was input into the part of uncertain rolling optimization, and the uncertain rolling OS was applied to optimize the economic dispatch in multiple uncertain environments. Moreover, the part of the historical data update could use the latest source-load data under the ROF to improve the prediction accuracy of the probability distribution by updating the historical data.

The prediction of electrical, cooling, and heating energies on different time scales is shown in Fig. 3. The prediction time interval of electrical energy (including RES generation and electrical load) is set as Δs , and the prediction time interval of cooling and heating energy is set as Δl . At time t , based on historical data, the probability distribution of electrical cooling and heating energies is predicted simultaneously. At time $t + \Delta s$, only the probability distribution of electric energy is predicted, and at time $t + \Delta l$, the probability distribution of electrical cooling and heating energies is predicted simultaneously. Similarly, in the part of historical data update, the time resolution of the electrical energy update historical data is Δs , and the resolution of the cooling and heating energy update historical data is Δl .

3.2. RMPC-based OS

Based on the integrated uncertainty ROF, an RMPC-based OS was proposed. The OS comprises two parts. During optimization, the probability distribution of the source-load prediction was determined first. Afterward, the equipment scheduling was optimized through a robust OS according to the probability distribution information to meet the supply and demand requirements. At the next time interval, the predicted probability distribution and robust optimization were re-conducted based on the latest source-load information. The flowchart of the proposed OS is shown in Fig. 4.

- 1) GPR-based source-load prediction: Predictions were performed with different time scales for RES and the load. In the prediction time horizon, the source-load probability distribution (predicted value and its corresponding error) can be realized by the GPR method according to the historical data.
- 2) Min-max rolling optimization: Min-max rolling optimization was performed within each time interval using the probability distribution information to minimize the operating costs under an uncertain scenario over the given optimization time horizon.

The flow of the proposed OS is summarized as follows:

Step 1: Set the prediction and rolling optimization time horizons.

Step 2: For each time interval, update the source-load data according to the latest state of the system and obtain the probability distribution of RES and the load forecast through the GPR-based source-load prediction.

Step 3: Obtain the source-load predicted value and corresponding error through the probability distribution.

Step 4: Perform the min-max rolling optimization with the prediction information from Step 3 and implement the optimization

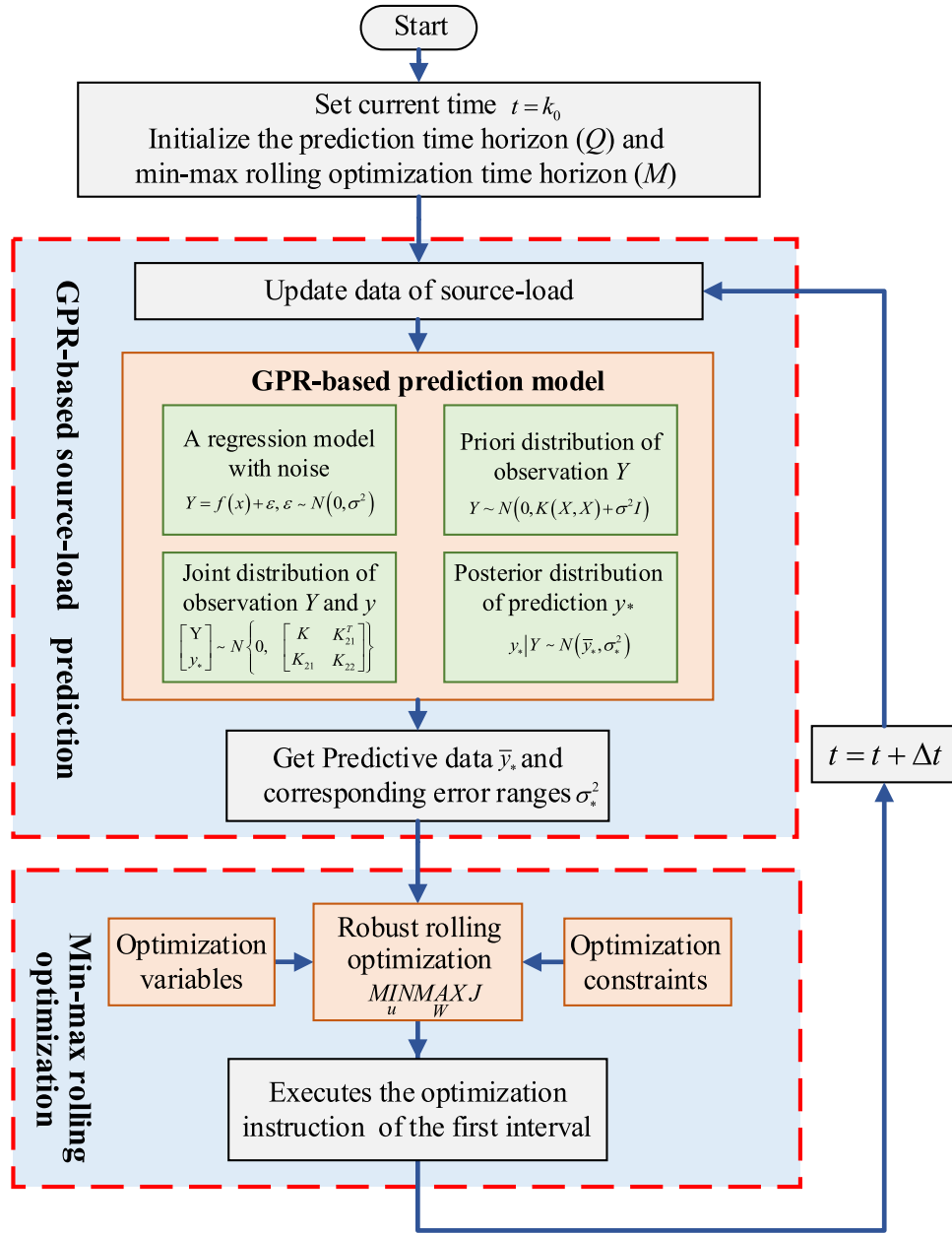


Fig. 4. Flowchart of RMPC-based OS.

instruction of the first interval.

Step 5: Update $t = t + \Delta t$ and return to Step 2.

3.2.1. GPR-based source-load prediction

The GPR is a nonparametric method based on the Bayesian theory and can be used to obtain predicted probability distributions. Consequently, the GPR method was used herein to obtain the probability distribution of the RES generation and load consumption forecast.

In the GPR-based source-load prediction, different time scales were used for electricity, cooling, and heating load forecasts. Specifically, the cooling and heating loads were predicted using data with hourly resolution. Meanwhile, the electrical load and RES power generation were predicted using data with a 15-min time resolution. Considering the RES prediction as an example, the detailed implementation of the GPR can be obtained as follows:

Given a wind power sequence $\{y_1, y_2, \dots\}$, the training set is assumed as $S = \{(x_i, y_i) \mid i = 1, 2, \dots, M\}$. The training set is constructed from the

original data, where $X_i = [y_{t-i}, y_{t-i-1}, \dots, y_{t-i-p+1}]^T$ represents the p -dimension input data, and $Y_i = y_{t-i+1}$ denotes the output data. It is assumed that the relationship between the input and output can be expressed as follows:

$$Y = f(X) + \varepsilon \quad (19)$$

where the noise ε obeys the normal distribution of $N(0, \sigma^2)$, and $f(x)$ is an underlying function defined by the Gaussian process, which can be expressed as follows:

$$f(X) \sim gp(m(X), k(X, X)) \quad (20)$$

Here $m(X)$ denotes the mean function, $k(X, X)$ is a kernel function that is selected to represent the covariance function. In most applications, the mean function $m(X)$ is set to 0. Consequently, from Eq. (19), the prior distribution of Y can be described as: $Y \sim N(0, k(X, X) + \sigma^2 I_n)$, where I_n represents an n -dimensional identity matrix.

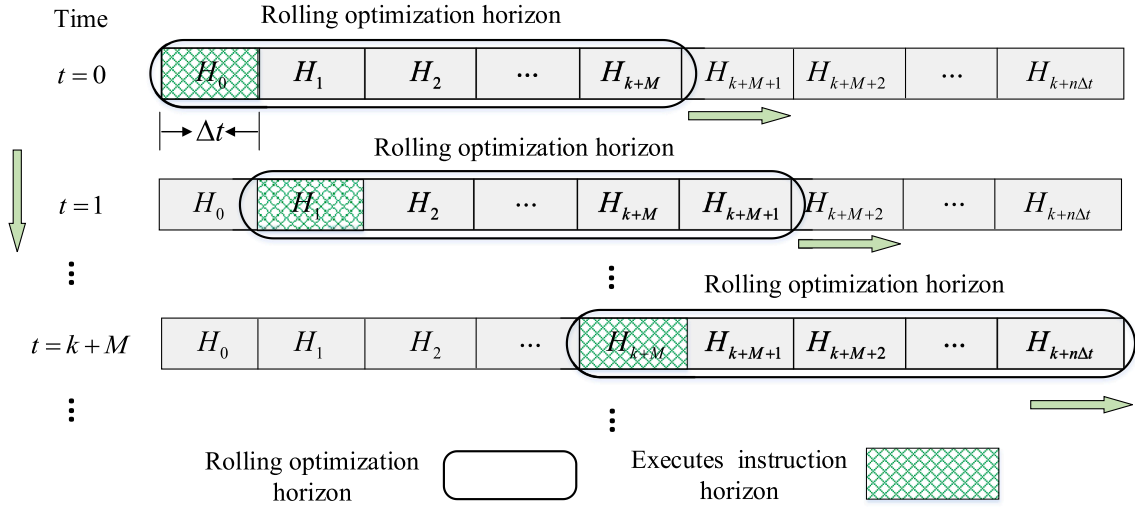


Fig. 5. Time sequences of min-max rolling optimization.

For $Y = [Y_i]_{i=1}^N, X = [X_i]_{i=1}^N$. To obtain the predicted value (y_*) of a new input data (x_*), the joint distribution of Y and y_* can be given by Eq. (21).

$$\begin{bmatrix} Y \\ y_* \end{bmatrix} \sim N \left(0, \begin{bmatrix} K(X, X) + \sigma^2 I_N & K(X, x_*) \\ K(x_*, X) & K(x_*, x_*) \end{bmatrix} \right) \quad (21)$$

where I_N is an $N \times N$ identity matrix, $K(X, X)$ represents the covariance matrix between the training set, $K(X, x_*) = K(x_*, X)^T$ represents the covariance matrix between the training set X and the test set x_* , and $K(x_*, x_*)$ denotes the covariance matrix of the test set itself. The covariance matrix can be calculated from the kernel function k . In Eq. (21), the covariance matrix can be described as follows:

$$K(X, X) \triangleq \begin{bmatrix} k(X_1, X_1) & k(X_1, X_2) & \cdots & k(X_1, X_N) \\ k(X_2, X_1) & k(X_2, X_2) & \cdots & k(X_2, X_N) \\ \vdots & \vdots & \ddots & \vdots \\ k(X_N, X_1) & k(X_N, X_2) & \cdots & k(X_N, X_N) \end{bmatrix} \quad (22)$$

$$K(X, x_*) = K(x_*, X)^T \triangleq [k(X_1, x_*), \dots, k(X_N, x_*)]^T \quad (23)$$

The square exponential kernel function is normally used in recent studies [13], defined by Eq. (24).

$$k(a, b) = \lambda_1^2 \exp \left(-\frac{(a-b)^2}{2\lambda_2^2} \right) \quad (24)$$

where λ_1 and λ_2 are the hyperparameters, and can be obtained by maximizing the log marginal likelihood in Eq. (25).

$$\log(Y|X, \lambda) = -\frac{N}{2} \log(2\pi) - \frac{1}{2} \log |K(X, X) + \sigma^2 I_N| - \frac{1}{2} Y^T (K(X, X) + \sigma^2 I_N)^{-1} Y \quad (25)$$

The probability distribution of the predicted value $y_*|Y$ can be described as $y_*|Y \sim N(\bar{y}_*, \sigma_*^2)$, where $\bar{y}_*$ and σ_*^2 can be calculated as follows:

$$\bar{y}_* = K(x_*, X) (K(X, X) + \sigma^2 I_N)^{-1} Y \quad (26)$$

$$\sigma_*^2 = K(x_*, x_*) - K(x_*, X) (K(X, X) + \sigma^2 I_N)^{-1} K(X, x_*) \quad (27)$$

The predicted value is $\bar{y}_*$, and the 95% confidence interval corresponding to the predicted value is $[\bar{y}_* - 1.96\sigma_*, \bar{y}_* + 1.96\sigma_*]$.

3.2.2. Min-max rolling optimization

According to the flowchart of the RMPC-based OS, after obtaining the distribution of the source-load probability predictions, a min-max rolling optimization model is proposed. The min-max rolling optimization is achieved based on the prediction information over a time interval $[k, k+M]$. However, only the instruction at $t=k$ is executed. Afterward, the rolling optimization horizon moves forward by one interval. The time sequences of the min-max rolling optimization are shown in Fig. 5.

- (1) Objective function – Herein, the operating cost is chosen as the objective function, and it only considers the energy consumption. The objective function can be expressed as follows:

$$\min_{U(t)} \max_{W(t)} J_t = \sum_{t=k+1}^{k+M} [P_{\text{gas}}(G_{\text{pgu}}(t) + G_{\text{gb}}(t)) + P_{\text{grid}}(t)E_{\text{grid}}(t)] \quad (28)$$

where $t = k+1, k+2, \dots, k+M$, P_{gas} represents the natural gas price, and $P_{\text{grid}}(t)$ represents the time-of-use electricity price. U , M and W represent the optimization variables, rolling optimization horizon, and error set of the predicted value, respectively.

- (2) Constrains – Besides satisfying the constraints of Eqs. (1)–(18), the optimization variables must satisfy the following capacity constraints of the equipment:

$$0 \leq E_{\text{pgu}}(t) \leq E_{\text{pgu}, \max} \quad (29)$$

$$0 \leq H_{\text{gb}}(t) \leq H_{\text{gb}, \max} \quad (30)$$

$$0 \leq C_{\text{ac}}(t) \leq C_{\text{ac}, \max} \quad (31)$$

$$0 \leq C_{\text{ec}}(t) \leq C_{\text{ec}, \max} \quad (32)$$

where $E_{\text{pgu}, \max}$, $H_{\text{gb}, \max}$, $C_{\text{ac}, \max}$ and $C_{\text{ec}, \max}$ represent the rated power values of PGU, GB, AC, and EC, respectively.

Herein, the COP of the AC and EC and the efficiency of the PGU and GB were simplified as fixed values. Therefore, the expressions of the objective and constraint functions are linear, and the min-max rolling optimization is a typical mixed-integer linear programming (MILP) problem and is described as follows:

$$\begin{aligned} \min_{U(t)} \max_{W(t)} J_t \\ \text{s.t. Eqs. (1) – (9), (14)} \\ \text{Eqs. (10) – (13), (15) – (18), (29) – (32).} \end{aligned} \quad (33)$$

Here J_t represents the objective function, Eqs. (1)–(9) and (14) represent the equality constraints, Eqs. (10)–(13), (15)–(18), and (29)–(32) represent the inequality constraints.

4. Solution of min-max rolling optimization model

The above min-max rolling optimization is difficult to solve directly and must be converted into a tractable form. Herein, considering the objective function in Eq. (28) and the energy flow balance constraints, the following equations can be obtained:

$$\min_{U(t)} \max_{W(t)} J_t = \sum_{t=k+1}^{k+M} P_{\text{gas}} \left(\frac{E_{\text{pgu}}(t)}{\eta_{\text{pgu}}} + \frac{H_{\text{gb}}(t)}{\eta_{\text{gb}}} \right) + P_{\text{grid}}(t) E_{\text{grid}}(t) \quad (34)$$

where E_{grid} and H_{gb} can be expressed as follows.

$$E_{\text{grid}}(t) = \frac{C_{\text{ec}}(t)}{COP_{\text{ec}}} + E_1(t) + E_1^{\text{err}}(t) - E_{\text{pgu}}(t) - E_{\text{es,dis}}(t) + E_{\text{es,chr}}(t) - E_{\text{res}}(t) - E_{\text{res}}^{\text{err}}(t) \quad (35)$$

$$H_{\text{gb}}(t) = \frac{H_1(t)}{\eta_{\text{he}}} + \frac{C_1(t) - C_{\text{ec}}(t) + C_{\text{err}}(t)}{COP_{\text{ac}}} - H_{\text{hr}}(t) - H_{\text{hs,dis}}(t) + H_{\text{hs,chr}}(t) + H_{\text{err}}(t) \quad (36)$$

$$\begin{aligned} E_{\text{sta}}(t+1) &= aE_{\text{sta}}(t) + E_{\text{es,chr}}(t)\eta_{\text{es,chr}} - E_{\text{es,dis}}(t)/\eta_{\text{es,dis}}, \\ E_{\text{sta}}(t+2) &= a^2E_{\text{sta}}(t) + aE_{\text{es,chr}}(t)\eta_{\text{es,chr}} + E_{\text{es,chr}}(t+1)\eta_{\text{es,chr}} - aE_{\text{es,dis}}(t)/\eta_{\text{es,dis}} - E_{\text{es,dis}}(t+1)/\eta_{\text{es,dis}}, \\ &\vdots, \\ E_{\text{sta}}(t+Q) &= a^Q E_{\text{sta}}(t) + a^{Q-1}E_{\text{es,chr}}(t)\eta_{\text{es,chr}} + \dots + E_{\text{es,chr}}(t+M-1)(a^{Q-M}\eta_{\text{es,chr}} + \dots + \eta_{\text{es,chr}}) \\ &\quad - a^{Q-1}E_{\text{es,dis}}(t)/\eta_{\text{es,dis}} - \dots - E_{\text{es,dis}}(t+M-1)(a^{Q-M}/\eta_{\text{es,dis}} + \dots + 1/\eta_{\text{es,dis}}). \end{aligned} \quad (44)$$

In Eq. (34), the set of state variables is assumed to be SV . The error set W is bounded, and other information is unknown. U , W , and SV can be expressed as:

$$U(t) = \{E_{\text{pgu}}(t), C_{\text{ec}}(t), H_{\text{hs}}(t), E_{\text{es}}(t)\} \quad (37)$$

$$W(t) = \{E_{\text{res}}^{\text{err}}(t), E_1^{\text{err}}(t), H_{\text{err}}(t), C_{\text{err}}(t)\} \quad (38)$$

$$SV(t) = \{E_1(t), E_{\text{res}}(t), H_1(t), C_1(t)\} \quad (39)$$

According to Eqs. (34)–(36), the optimization variables are $E_{\text{pgu}}(t)$, $C_{\text{ec}}(t)$, $H_{\text{hs}}(t)$, and $E_{\text{es}}(t)$.

In Eq. (38), the elements in the error set $W(t)$ satisfy the following constraints:

$$\begin{aligned} E_{\text{res}}^{\text{err}}(t) &= \delta_{\text{E, res}}^+(t) \bar{E}_{\text{res}}^{\text{err}}(t) - \delta_{\text{E, res}}^-(t) \underline{E}_{\text{res}}^{\text{err}}(t), \\ \sum_{t=k+1}^{k+M} \left(\delta_{\text{E, res}}^+(t) + \delta_{\text{E, res}}^-(t) \leq \prod_{\text{E, res}} \right), \quad 0 \leq \delta_{\text{E, res}}^+(t), \delta_{\text{E, res}}^-(t) \leq 1. \end{aligned} \quad (40)$$

$$\begin{aligned} E_1^{\text{err}}(t) &= \delta_{\text{E, 1}}^+(t) \bar{E}_1^{\text{err}}(t) - \delta_{\text{E, 1}}^-(t) \underline{E}_1^{\text{err}}(t), \\ \sum_{t=k+1}^{k+M} \left(\delta_{\text{E, 1}}^+(t) + \delta_{\text{E, 1}}^-(t) \leq \prod_{\text{E, 1}} \right), \quad 0 \leq \delta_{\text{E, 1}}^+(t), \delta_{\text{E, 1}}^-(t) \leq 1. \end{aligned} \quad (41)$$

$$\begin{aligned} H_{\text{err}}(t) &= \delta_{\text{H}}^+(t) \bar{H}_{\text{err}}(t) - \delta_{\text{H}}^-(t) \underline{H}_{\text{err}}(t), \\ \sum_{t=k+1}^{k+M} \left(\delta_{\text{H}}^+(t) + \delta_{\text{H}}^-(t) \leq \prod_{\text{H}} \right), \quad 0 \leq \delta_{\text{H}}^+(t), \delta_{\text{H}}^-(t) \leq 1. \end{aligned} \quad (42)$$

$$\begin{aligned} C_{\text{err}}(t) &= \delta_{\text{C}}^+(t) \bar{C}_{\text{err}}(t) - \delta_{\text{C}}^-(t) \underline{C}_{\text{err}}(t), \\ \sum_{t=k+1}^{k+M} \left(\delta_{\text{C}}^+(t) + \delta_{\text{C}}^-(t) \leq \prod_{\text{C}} \right), \quad 0 \leq \delta_{\text{C}}^+(t), \delta_{\text{C}}^-(t) \leq 1. \end{aligned} \quad (43)$$

The constraints in Eqs. (40)–(43) describe the uncertainty interval of

the RES generation and load consumption prediction error. In Eq. (36), $\bar{E}_{\text{res}}^{\text{err}}(t)$ and $\underline{E}_{\text{res}}^{\text{err}}(t)$ denote the upper and lower deviations of the RES generation prediction errors, respectively. $\delta_{\text{E, res}}^+(t)$ and $\delta_{\text{E, res}}^-(t)$ restrict $E_{\text{res}}^{\text{err}}(t)$ to the interval $[\bar{E}_{\text{res}}^{\text{err}}(t), -\underline{E}_{\text{res}}^{\text{err}}(t)]$. The parameter $\prod_{\text{E, res}} (0 \leq \prod_{\text{E, res}} \leq M)$ is called the uncertainty budget, which can adjust the conservatism of robust optimization. Specifically, if $\prod_{\text{E, res}} = 0$, there is no prediction error in the RES generation. If $\prod_{\text{E, res}} = M$, the prediction error of the RES generation can reach its lower or upper limit throughout all optimization horizons. The definitions of symbols and variables in Eqs. (41)–(43) are similar to those in Eq. (40).

Energy storage units have continuous characteristics with time, and the state of energy storage at each time step affects the state and decision-making of the next time step. Consequently, the multi-time dynamic characteristics must be considered in the MPC-based rolling optimization process and the state-space model is usually used to establish the dynamics of energy storage units in the application of the MPC case [35]. ESU and the HSU have similar dynamic characteristics. Considering the ESU, for example, the Eq. (44) can be obtained by expanding the Eq. (9) in chronological order.

where $a = (1 - \tau_{\text{es}})$, Q is the prediction time horizon, and M is the optimization time horizon. The Eq. (44) can be expressed as follows:

$$\bar{E}_{\text{sta}}(t) = A E_{\text{sta}}(t) + G_{\text{es,chr}} \bar{E}_{\text{es,chr}}(t) - G_{\text{es,dis}} \bar{E}_{\text{es,dis}}(t) \quad (45)$$

$$\bar{E}_{\text{sta}}(t) = \begin{bmatrix} E_{\text{sta}}(t+1) \\ \vdots \\ E_{\text{sta}}(t+Q) \end{bmatrix} \bar{E}_{\text{es,chr}}(t) = \begin{bmatrix} E_{\text{es,chr}}(t) \\ \vdots \\ E_{\text{es,chr}}(t+M-1) \end{bmatrix}$$

$$A = \begin{bmatrix} a \\ \vdots \\ a^Q \end{bmatrix} G_{\text{es,chr}} = \begin{bmatrix} \eta_{\text{es,chr}} & 0 & 0 \\ \vdots & \vdots & 0 \\ a^{M-1}\eta_{\text{es,chr}} & \dots & \eta_{\text{es,chr}} \\ \vdots & \vdots & \\ a^{Q-1}\eta_{\text{es,chr}} & \dots & \sum_{i=0}^{Q-M} a^i \eta_{\text{es,chr}} \end{bmatrix}$$

The RMPC-based closed-loop min-max optimization was adopted herein. It is assumed that the future control U_{t+1} can be optimally calculated when the future state SV_{t+1} is available. Consequently, the problem to be solved in the closed-loop min-max rolling optimization can be expressed as follows:

$$\begin{aligned} U &= \min_{u_t} \max_{w_t} \dots \min_{u_{t+M-1}} \max_{w_{t+M-1}} J_t \\ J_t &= \zeta(SV_t, U_t, W_t, \dots, SV_{t+M-1}, U_{t+M-1}, W_{t+M-1}). \end{aligned} \quad (46)$$

To solve the above min-max rolling optimization problem, the UB is defined as the upper bound of J_t , for all disturbances $W_t^{(i)}$. When the constraint $J_t \leq \text{UB}$ is satisfied, the determined corresponding optimization decision is $U_t^{(i)}$. From Eq. (33), the optimization variables $U_t^{(i)}$ and the related state realization $SV_t^{(i)}$ are introduced for each possible disturbance realization $W_t^{(i)}$, where i represents the index of all possible disturbance realizations, and $j = 1, 2, 3, 4$ represents the index of the

Table 1
Time-of-use electricity price.

Period time	Detailed time	Electricity price [CNY/(kWh)]
Peak	8:00–11:00 18:00–23:00	1.07
Intermediate	7:00–8:00 11:00–18:00	0.68
Lowest	23:00–7:00	0.26

Table 2
Equipment parameter value.

Component	Symbol	Value
PGU	$E_{pgu,max}$	300 kW
GB	$H_{b,max}$	400 kW
AC	$C_{ab,max}$	550 kW
EC	$C_{ec,max}$	550 kW
ESU	$E_{sta,max}$	750 kWh
HSU	$H_{sta,max}$	600 kWh

optimization variable. Then, the optimization model can be transformed into Eqs. (47)–(56).

$$\min_{UB, U_t^{(i)}} UB \quad (47)$$

$$s.t. \quad \sum_{t=k+1}^{k+M} P_{gas} \left(\frac{E_{pgu}^{(i)}(t)}{\eta_{pgu}} + \frac{H_{gb}^{(i)}(t)}{\eta_{gb}} \right) + P_{grid}(t) E_{grid}^{(i)}(t) \leq UB \quad (48)$$

$$0 \leq \frac{H_i(t)}{\eta_{he}} + \frac{C_i(t) - C_{ec}^{(i)}(t) + C_{err}^{(i)}(t)}{COP_{ac}} - H_{hr}^{(i)}(t) - H_{hs}^{(i)}(t) + H_{err}^{(i)}(t) \leq H_{gb,max} \quad (49)$$

$$0 \leq C_i(t) - C_{ec}^{(i)}(t) + C_{err}^{(i)}(t) \leq C_{ac,max} \quad (50)$$

$$\forall E_{res}^{(i)} \in [\delta_{E,res}^+ \bar{E}_{res}^{err}, \delta_{E,res}^- \underline{E}_{res}^{err}], \forall E_{err}^{(i)} \in [\delta_{E,l}^+ \bar{E}_l^{err}, \delta_{E,l}^- \underline{E}_l^{err}] \quad (51)$$

$$\forall C_{err}^{(i)} \in [\delta_C^+ \bar{C}_{err}(t), \delta_C^- \underline{C}_{err}(t)], \forall H_{err}^{(i)} \in [\delta_H^+ \bar{H}_{err}, \delta_H^- \underline{H}_{err}]$$

Eqs. (40) – (43)

$$0 \leq \bar{E}_{es}^{(i)}(t) \leq \bar{E}_{es,max} \quad (52)$$

$$0 \leq \bar{H}_{hs}^{(i)}(t) \leq \bar{H}_{hs,max} \quad (53)$$

$$E_{sta,min} \leq A E_{sta}^{(i)}(t) - G_{es} \bar{E}_{es}^{(i)}(t) \leq E_{sta,max} \quad (54)$$

$$H_{sta,min} \leq B H_{sta}^{(i)}(t) - G_{hs} \bar{H}_{hs}^{(i)}(t) \leq H_{sta,max} \quad (55)$$

$$0 \leq U_{jt}^{(i)} \leq U_{j,max} \quad (56)$$

In the optimization model described by Eqs. (47)–(56), all the realizations of the error set need to be considered (from Eq. (51)), which is a difficult problem to solve. However, it can be seen from Ref. [36] that in this case the control must be optimized only when an extreme disturbance is realized. Therefore, the min–max optimization model can be solved by considering only the boundary conditions of the error set, indicating that the vertices of $W_t^{(i)}$ can be selected to get the worst case max J_t . As a result, the optimization model can be transformed into the form of Eq. (57).

Table 3
Equipment performance parameters.

Parameters	Symbol	Value	Sources
Electrical efficiency of PGU	η_{pgu}	0.3	[34]
COP of AC	COP_{ac}	1.2	[34]
COP of EC	COP_{ec}	3.0	[37]
Efficiency of GB	η_{gb}	0.9	[37]
Efficiency of HR	η_{hr}	0.75	[34]
Efficiency of HE	η_{he}	0.98	[37]
Charge/discharge efficiency of ESU	η_{es}	0.98	[34]
Self-discharge coefficient of ESU	τ_{es}	0.0002	Experiment
Charge/discharge efficiency of HSU	η_{hs}	0.9	[34]
Self-discharge coefficient of HSU	τ_{hs}	0.005	[38]

$$\min_{UB, U_t^{(i)}} UB$$

$$s.t. \quad \sum_{t=k+1}^{k+M} P_{gas} \left(\frac{E_{pgu}^{(i)}(t)}{\eta_{pgu}} + \frac{H_{gb}^{(i)}(t)}{\eta_{gb}} \right) + P_{grid}(t) E_{grid}^{(i)}(t) \leq UB$$

$$0 \leq \frac{H_i(t)}{\eta_{he}} + \frac{C_i(t) - C_{ec}^{(i)}(t) + C_{err}^{(i)}(t)}{COP_{ac}} - H_{hr}^{(i)}(t) - H_{hs}^{(i)}(t) + H_{err}^{(i)}(t) \leq H_{gb,max}$$

$$0 \leq C_i(t) - C_{ec}^{(i)}(t) + C_{err}^{(i)}(t) \leq C_{ac,max}$$

$$E_{res}^{(i)} \in \{ \delta_{E,res}^+ \bar{E}_{res}^{err}, \delta_{E,res}^- \underline{E}_{res}^{err} \}, E_{err}^{(i)} \in \{ \delta_{E,l}^+ \bar{E}_l^{err}, \delta_{E,l}^- \underline{E}_l^{err} \},$$

$$C_{err}^{(i)} \in \{ \delta_C^+ \bar{C}_{err}(t), \delta_C^- \underline{C}_{err}(t) \}, H_{err}^{(i)} \in \{ \delta_H^+ \bar{H}_{err}, \delta_H^- \underline{H}_{err} \}$$

Eqs. (40) – (43), Eqs. (52) – (56)

(57)

In Eq. (57), the min–max rolling optimization model is transformed into a tractable form by picking the boundary values of the error set. Eq. (57) reveals this problem is a MILP problem, and the min–max rolling optimization model can be solved easily by a commercial solver.

5. Case studies

The RMPC-based optimization proposed in the study is general and valid regardless of the system changes. Therefore, to verify the effectiveness of the proposed method, a hypothetical RES-CCHP building was used for the case studies. The system structure is constructed as shown in Fig. 1; wind power is used as the renewable energy in the system. In rolling optimization, the time interval (Δt) was set as 15 min, and the time horizon was set to be 1 h, resulting in four intervals for each rolling time horizon.

The case studies were conducted based on a typical summer day, and the system was not allowed to sell electricity to the MG. Table 1 shows the time-of-use electricity price. Herein, the heating value of the natural gas is 35.59 MJ/m³ and the natural gas price is 3.15 CNY/m³. The equipment parameter value of the RES-CCHP system components are shown in Table 2, and Table 3 lists the parameters used and their associated sources.

The case studies were coded in MATLAB R2018b using YALMIP for the min–max rolling optimization and GPML for the source-load uncertainty prediction based on GPR. At each optimization horizon, GPML makes predictions and feeds them to the YALMIP model for optimization. The program was run on an Intel(R) Core™ i7-9700 3.00 GHz personal computer with 8 GB RAM.

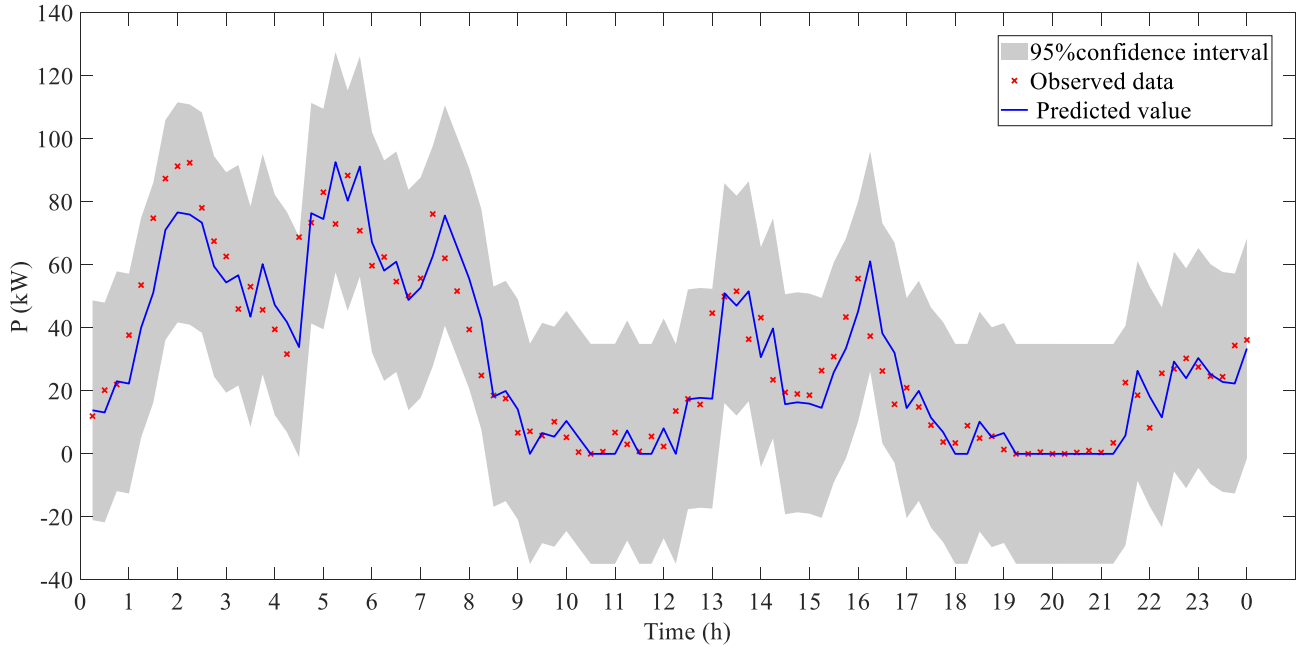


Fig. 6. Observed and predicted wind power.

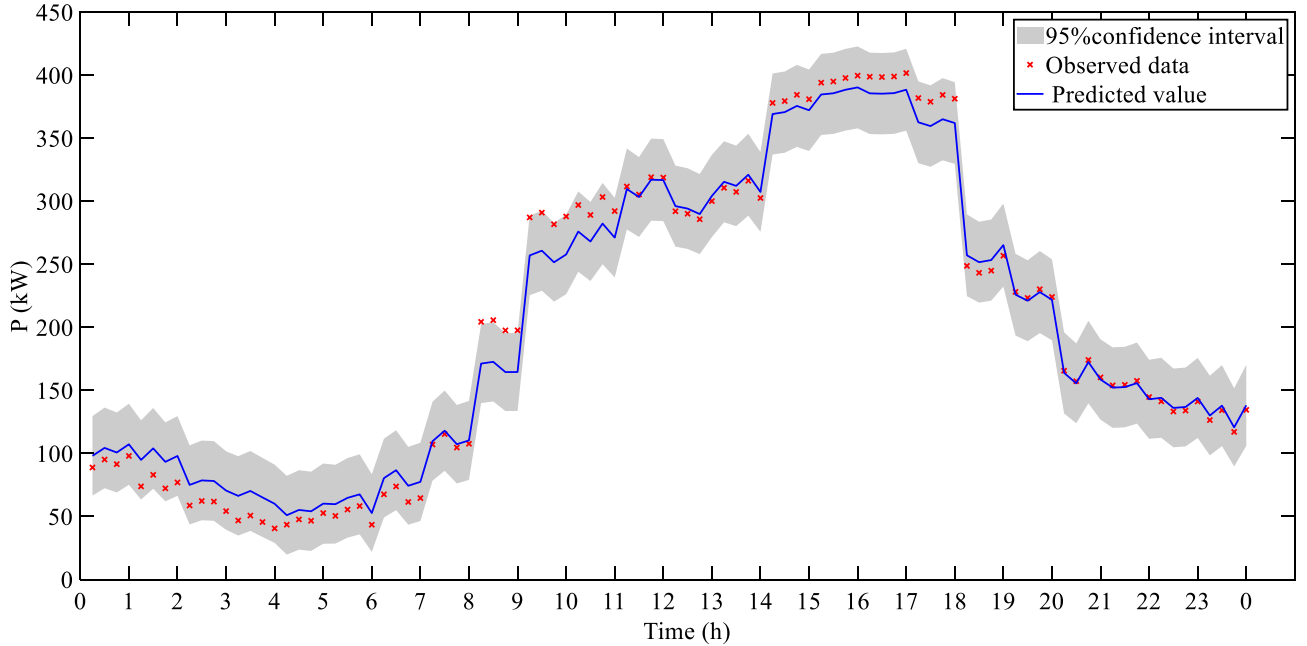


Fig. 7. Observed and predicted electrical load.

5.1. GPR-based data prediction

The proposed GPR method achieved the probability distribution predictions of the wind power, electrical load, and cooling load for a typical summer day. Considering the long timescale of cooling and heating loads, they were assumed to remain the same within one hour.

The predicted value and its confidence interval are shown in Figs. 6–8. For practical engineering considerations, if the wind power predicted value is negative, it is automatically set to zero. To facilitate the analysis of prediction data, the evaluation metrics of point prediction and interval prediction are introduced.

Point prediction evaluation metric: coefficient of determination (R^2).

R^2 represents the ratio of the sum of squares of the regression to the sum of squares of the total deviations. The closer the value of R^2 is to 1, more accurate is the prediction. Interval prediction evaluation metric: coverage probability (C_α). C_α is expressed as the probability that the observed value falls within the prediction interval at the α confidence level [39].

R^2 and C_α can be calculated using Eq. (58), where $\bar{y}_{s,a}$, Y_a and $\bar{Y}$ represent the predicted value, observed value, and mean of the observed values, respectively; P_s is the size of the predicted data sample, and c_α is the number of samples in the prediction interval.

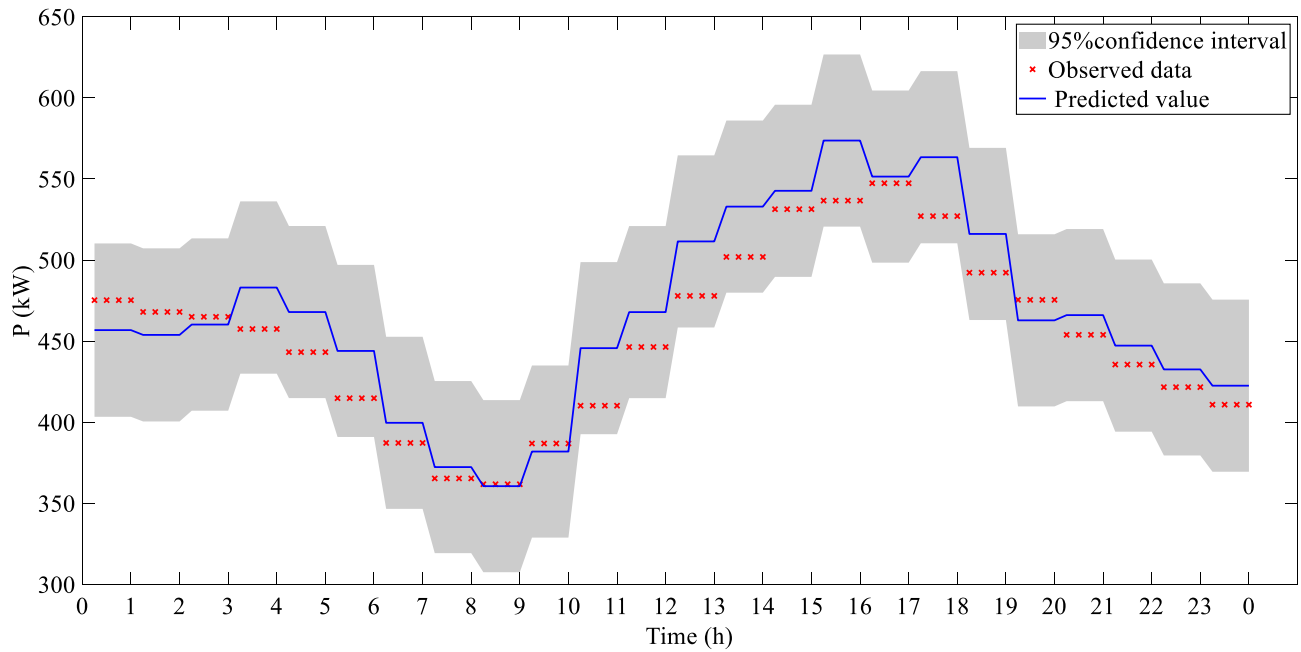


Fig. 8. Observed and predicted cooling load.

Table 4
Point prediction and interval prediction metrics for forecasting data.

Items	Metrics	
	R^2	C_a
Wind power	0.843	100%
Electrical load	0.986	95.8%
Cooling load	0.833	100%

$$R^2 = 1 - \frac{\sum_{a=1}^{P_s} (\bar{y}_{s,a} - Y_a)^2}{\sum_{a=1}^{P_s} (Y_a - \bar{Y})^2} C_a = \frac{c_a}{P_s} \times 100\% \quad (58)$$

The metrics of the prediction data are shown in Table 4. The R^2 -metric for electrical load prediction is higher than that for wind energy and cooling load, indicating that the point prediction accuracy of electrical load is high. The wind power and cooling load prediction data have a C_a -metric of 100%, indicating that all their observed data fall within the 95% confidence interval. Although, the observed data of

electrical load do not all fall within the 95% confidence interval, its C_a -metric is $> 90\%$. It is shown that the source-load probability distribution can be used for RMPC-based optimization.

5.2. Comparison and analysis of different energy storage unit scheduling methods

The uncertainty budgets $\Pi_{E, \text{res}} = \Pi_{E,1} = \Pi_H = \Pi_C = 2$ were selected for the following comparative case studies. Fig. 9 shows the optimal electricity dispatch result by the RMPC-based OS with the rolling dispatch of all equipment. Thus, it can be seen that all WT is consumed as the WT incurs no energy costs. The PGU provides most of its electricity during peak electricity prices. When the price of electricity is at its lowest, electricity is provided primarily by the MG. The ESU can reduce costs by charging and discharging when the electricity price is low and high, respectively. However, evidently, the ESU has been ineffectively used because the time horizon of each rolling optimization is short.

To make full use of the advantages of the ESU, during RMPC-based

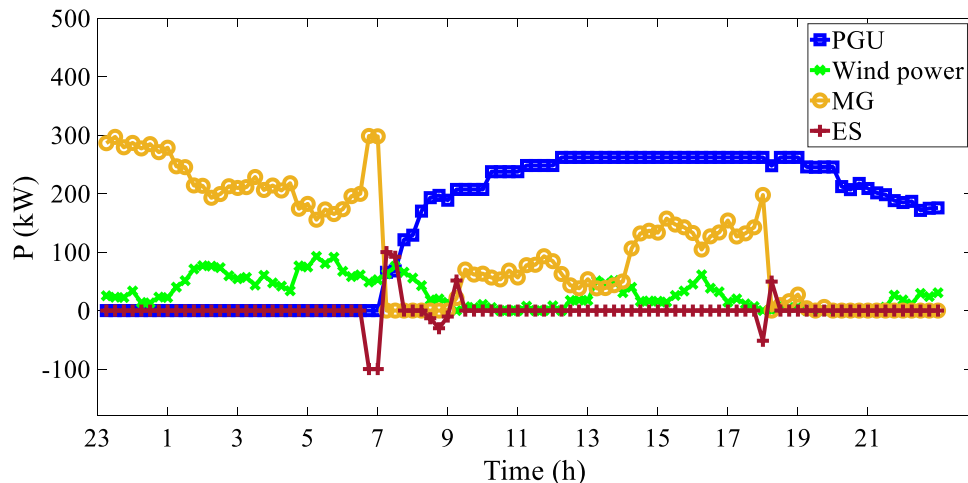


Fig. 9. RMPC-based optimal electricity dispatch results in rolling dispatch for all equipment.

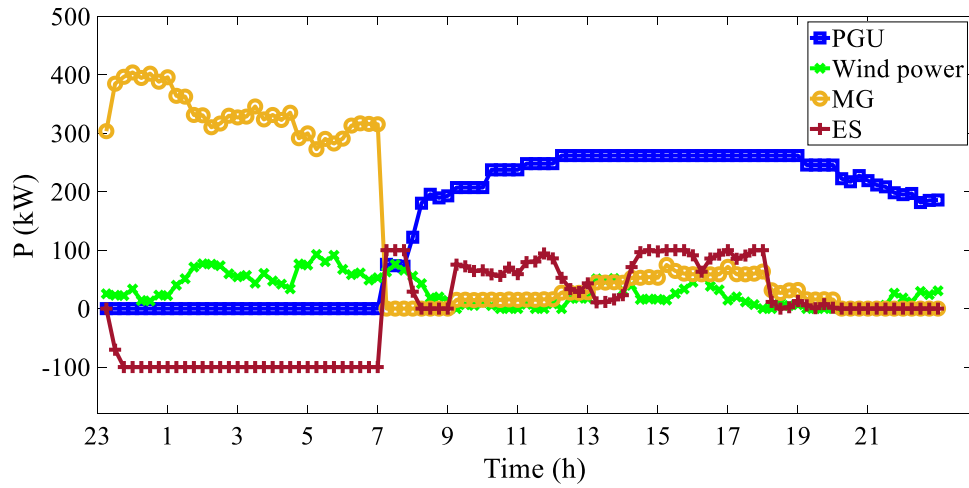


Fig. 10. RMPC-based optimal electricity dispatch combined with day-ahead scheduling of ES.

Table 5

Operating costs of optimization with or without day-ahead optimization scheduling of ESU.

Optimized strategy	RMPC-based optimization with rolling dispatch of all equipment	RMPC-based optimization with day-ahead scheduling of ESU
Cost (CNY)	19,941	17,518

Table 6

Computational time for each time interval of OS.

Computational time	GPR-based prediction	RMPC-based optimization	Total time (15 min)
Max (s)	6.5772	2.1167	8.6939

optimization, ESU adopts the scheduling determined by the day-ahead optimization. In day-ahead scheduling, the optimization objective is the operating costs, and the optimization constraints are system energy balance and equipment performance constraints, including Eqs. (1)–(14) and (25)–(28).

Fig. 10 shows the optimal power dispatch result by the RMPC-based OS combined with day-ahead scheduling of ESU. Table 5 presents the

Table 7

Operating costs of different OSs.

Optimized algorithm	RMPC-based optimization	TMPC-based optimization
Operating costs (CNY)	17,518	19,789

operating costs with or without day-ahead optimization scheduling of ESU. The ESU is fully charged when the electricity price is at its lowest, and discharged when the electricity price is high, further reducing operating costs. From Table 5, the operating cost of RMPC-based optimization with day-ahead scheduling of ESU is 12.2% lower than that with only rolling dispatch of all equipment. Table 6 presents the computational time for each interval of the OS. The total maximum computational time of this OS is less than 15 min, which can ensure that the optimal scheduling results can be obtained within the scheduling interval. Due to the advantages of the day-ahead scheduling method in developing the ESU dispatch, in all subsequent RMPC optimization simulations, the ESU dispatch is given by the day-ahead scheduling method.

Fig. 11 presents the optimal cooling dispatch result by the RMPC-based OS. The blue and orange bars show the cooling produced by AC and EC devices, respectively. The green line represents the cooling load. When the electricity price culminates and is lowest, the cooling load is provided mainly by the AC and EC devices, respectively.

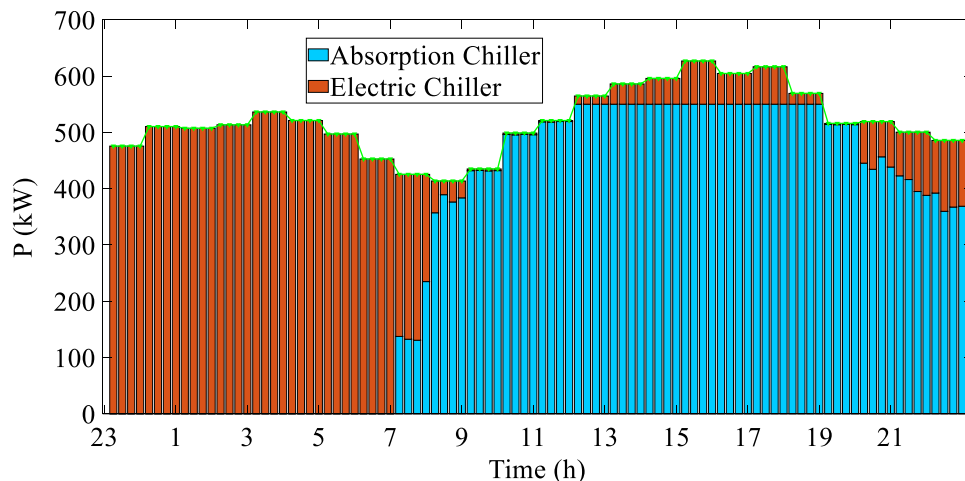


Fig. 11. RMPC-based optimal cooling dispatch combined with day-ahead scheduling of ESU.

Table 8

Maximum and minimum values of error range.

Item	Minimum (kW)	Maximum (kW)	Deviation between maximum and minimum (kW)
RES error	8.69	8.73	0.04
Electrical load error	30.64	32.99	2.35
Heating load error	20.41	20.52	0.11
Cooling load error	53.05	53.45	0.4

Table 9

Operating costs of robust optimization in different ways.

Optimized strategy	RMPC-based optimization	Day-ahead robust optimization
Cost (CNY)	19,419	19,694

5.3. Comparison with traditional MPC OS

To verify the effectiveness and superiority of the proposed OS, the RMPC-based OS was compared to a traditional MPC-based (TMPC-based) OS. The TMPC-based OS does not consider the existence of the prediction error, thereby outperforming the RMPC-based OS in terms of operating economy. However, it lacks robustness in system operation. With uncertain source-loads, the scheduling strategy, determined based on TMPC, deviates from the actual load, and these imbalances must be compensated by the MG, which increases the TMPC-based OS cost. Concerning the uncertainty budgets $\prod_{E, \text{res}} = \prod_{E, l} = \prod_H = \prod_C = 2$, the operating costs of these two OSs are shown in Table 7.

Table 7 shows that under the uncertain scenarios, the operating costs of the TMPC-based OS are high due to the compensation costs. However, the RMPC-based OS considers the source-load uncertainty, and is robust in the scheduling strategy. Compared with the TMPC-based OS, the operating costs are reduced by 11.5%.

5.4. Comparison with day-ahead robust OS

The day-ahead robust OS formulates the scheduling strategy for the whole day of the next day in an uncertain scenario, and the source-load information of the second day cannot be applied. In the RMPC-based optimization, prediction and optimization are implemented under the uncertainty ROF. The source-load information of the next day can be applied to reduce forecast deviation and system operating costs. In the day-ahead robust optimization, the maximum error obtained by the GPR-based PM is selected as its uncertainty scenario to cope with the

source-load uncertainty of the next day. To better prove the advantages of the RMPC-based OS, the worst scenario $\prod_{E, \text{res}} = \prod_{E, l} = \prod_H = \prod_C = M$ is selected to compare the economics of the two OSs. Table 8 presents the maximum and minimum values of the prediction error range by the GPR, and Table 9 presents the operating costs of these two OSs.

Table 9 shows that although the deviation between the maximum and minimum error values is minimal, the cost of the RMPC-based optimization is marginally lower than that of day-ahead robust optimization. These results reflect that the economic performance of an RMPC-based OS under ROF is better than a day-ahead robust OS.

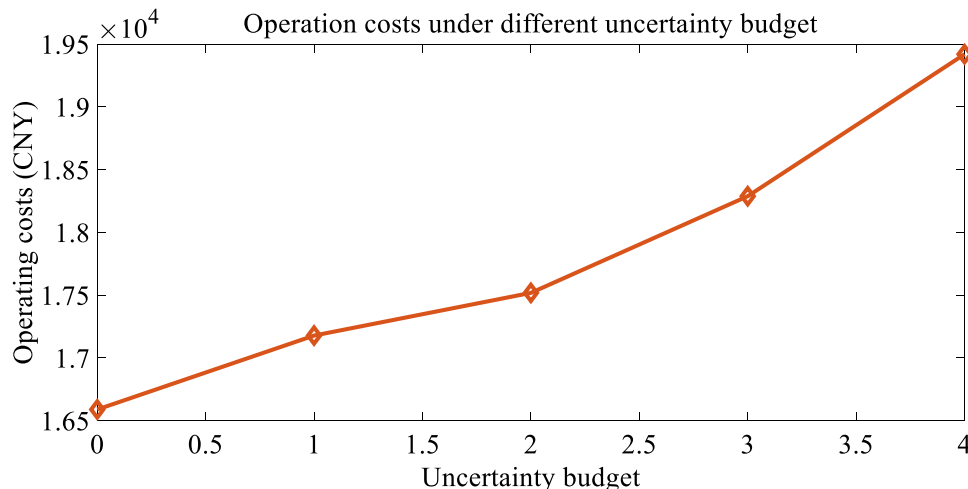
5.5. Sensitivity analysis of uncertainty budgets

In the RMPC-based OS, the uncertainty budgets $\prod_{E, \text{res}}, \prod_{E, l}, \prod_H, \prod_C$ determine the uncertainty level of the source-load. Here, the source-load uncertainty budgets are defined as the same value to analyze the uncertainty-level impact on the operating costs. The sensitivity of operating costs to the uncertainty budget is shown in Fig. 12. The RMPC-based OS proposed herein can guarantee the robustness and economy of the solution simultaneously when the uncertainty budget is appropriately selected.

6. Conclusion

Because of the multiple uncertainties of RES-CCHP systems, an integrated uncertainty ROF is established and an OS based on an RMPC is proposed. Case studies are used to verify the effectiveness and superiority of the proposed framework and OS. The following conclusions can be summarized:

- (1). In the source-load uncertainty scenario, the TMPC-based OS does not consider the source-load uncertainty, and the resulting compensation cost increases the total operating cost. The RMPC-based OS can obtain a robust system scheduling plan, reducing the operating cost by 11.5% compared with the TMPC-based OS.

**Fig. 12.** Operating costs at different uncertainty levels.

- (2). The OS proposed herein is implemented under the uncertainty ROF. The latest state of the system updates the historical data to obtain a more accurate source-load probability distribution. Compared with the day-ahead robust OS, the RMPC-based optimization exhibits better economic performance.
- (3). The uncertainty budget parameter introduction can adjust the conservatism of the robust optimization and increase the operational flexibility of the engineer. Moreover, the energy storage units adopt the day-ahead scheduling strategy, which can better exert advantages and reduce system operating costs.

CRediT authorship contribution statement

Xing Dong: Conceptualization, Methodology, Software, Writing – original draft. **Chenghui Zhang:** Conceptualization, Funding acquisition, Writing – review & editing. **Bo Sun:** Data curation, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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